

Xinchen Yao

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Education

Ph.D. Program (Starting September 2026):

The Chinese University of Hong Kong, Hong Kong, China
Ph.D. in Mechanical and Automation Engineering
Advisor: **Prof. Zhongyu Li**

Sep 2026 – Expected 2030

Dual-Degree Bachelor Program:

University of Illinois Urbana-Champaign, Urbana, Illinois, US
BS in Computer Engineering, GPA: 3.77/4.0

Sep 2022 – May 2026

Zhejiang University, Zhejiang, China

Sep 2022 – May 2026

BS in Electronic and Computer Engineering, GPA: 3.83/4.3

Publications

JOURNAL

Where to Learn: Analytical Policy Gradient Directed Exploration for On-Policy Robotic Reinforcement Learning [wheretolearn.github.io]

Leixin Chang, Xinchen Yao, Ben Liu, Liangjing Yang, Hua Chen

— *IEEE RA-L*, June 2026

World Translation: Minimizing Sim-to-Real Gap with Backward Dynamics Extraction and Unpaired Domain Translation

Xinchen Yao, Leixin Chang, Hua Chen

— *IEEE RA-L* (In preparation)

Technical Skills

Programming: Python, C/C++, CUDA, Matlab, Rust

Machine Learning: PyTorch, Jax, Reinforcement Learning, Generative Models

Robotics: Robot Simulators, Control Theory, ROS/ROS2, Computer Vision, SLAM

Systems: Linux, Embedded Systems, Parallel Computing

Awards and Honors

- Outstanding Student in Academic Achievement, ZJU-UIUC Institute (2024 - 2025)
- Zhejiang University Scholarship (2024 - 2025)
- Second Prize in RoboMaster University League Competition (2023, 2024)

Experience

Physical Intelligence Lab, Zhejiang, China

Undergraduate Researcher

Zhejiang University - University of Illinois Urbana-Champaign Institute

Jul 2025 – Jun 2026

Advisor: **Prof. Hua Chen**

- Developed locomotion controllers for bipedal robots and humanoids using reinforcement learning, achieving emergent behaviors that adapt to various terrains and conditions.
- Built evaluation pipelines to test trained robot policies across multiple simulation environments and deployed them to physical robots.
- Researched methods to reduce the performance gap between simulated and real-world robot operation through learned dynamics models.
- Improved reinforcement learning algorithms to achieve better training efficiency and performance stability.

Human Dynamics and Controls Lab, Illinois, US

Undergraduate Researcher

The Grainger College of Engineering, University of Illinois Urbana-Champaign

Sep 2024 – Jun 2025

Advisor: **Prof. Elizabeth Hsiao-Weckslar**

- Developed a dynamics model for a self-balancing ballbot platform, simulating omniwheel behavior while

maintaining computational efficiency.

- Implemented learning-based control to optimize balancing and movement performance and robustness.
- Integrated new force sensors into robot hardware, improving state estimation and control reliability.

RoboMaster Meta Team, Zhejiang, China

Zhejiang University - University of Illinois Urbana-Champaign Institute

Advisor: **Prof. Jiahuang Cui**

Control Group Leader

Jun 2023 – Jun 2026

- Led control group to win second prize in RoboMaster University League competition, managing a team developing software for multiple competitive robots.
- Designed and implemented a distributed control system supporting real-time coordination for multiple robots with modular architecture.
- Developed software for motor control, sensor integration, autonomous navigation, and decision-making algorithms.
- Trained and mentored new team members on control theory, robot software development, and algorithm design.
- Collaborated with mechanical team to optimize robot design for improved controllability and performance.

Projects

Omni WBR: Learning Adaptive Hybrid Wheeled-Biped Robot Omnidirectional Locomotion via Position-Based Incentive

First Author

- Developed a task design to generate adaptive omnidirectional gaits for wheeled bipedal robots, enabling smooth movement on uneven terrains without explicit gait planning.
- Implemented a training and deploying pipeline to evaluate the method in Isaac Lab simulator and physical robot.
- Demonstration available on my website: yao-xinchen.github.io/projects/omni-wbr

Meta-Team/Meta-ROS: Modular ROS2 Control Framework for multiple RoboMaster Competition Robots

Creator, Maintainer

- Created a comprehensive robot control system in ROS2 supporting multiple competitive robots with sensor integration, motion control, and autonomous coordination.
- Designed modular architecture enabling easy extension and real-time performance for competition scenarios.
- Integrated advanced capabilities including visual recognition, trajectory prediction, localization, and learning-based controllers.
- Code: github.com/Meta-Team/Meta-ROS

Custom Wheeled Biped Robot with RL-based Locomotion Controller

Leader

- Co-designed and built a custom wheeled biped robot with 4 degrees of freedom and closed chain structure, including low-level control system on embedded platform.
- Trained a reinforcement learning-based locomotion controller and deployed it on the physical robot.
- Demonstration available on my website: yao-xinchen.github.io/projects/wheeled-biped